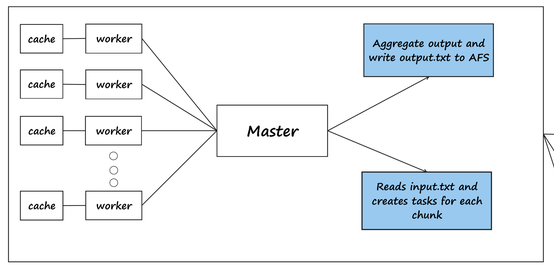
Application

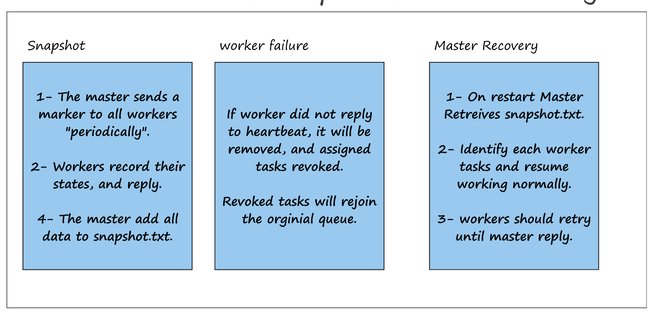
workers will read from AFS by RPC, with whole file caching functionality

The worker continuously requests tasks from the coordinator.

Worker retries registration, task requests, and result submissions up to 5 times.

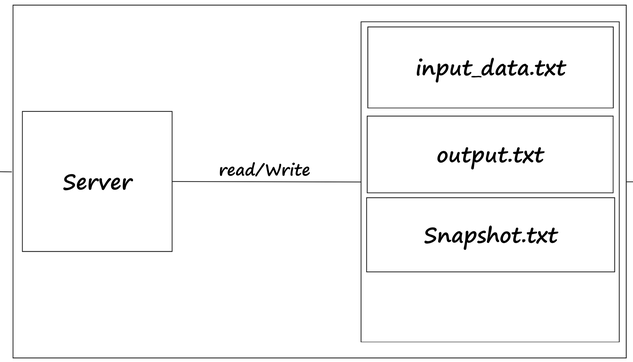


Failure Detection, Snapshot, and Recovery



AFS Active-Passive model

AFS - Primary



Replication

Primary replicate files to backup

Leader Election

1- Backup nodes wait for heartbeats from the Primary.

2- If no heartbeat arrives within a randomized timeout, the backup becomes a Candidate.

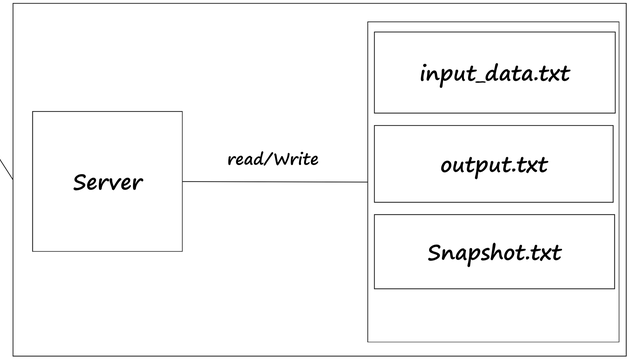
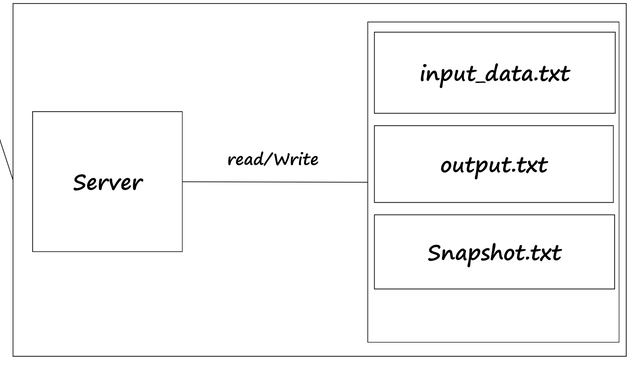
3- Candidate increments its term, votes for itself, and requests votes from peers.

4- If it gets a majority, it becomes Primary.

5- The new Primary sends periodic heartbeats to maintain authority.

6- If any node sees a higher term in a heartbeat or vote request, it steps down.

AFS - backup



AFS - backup